

Online Resource 1

AC-MOT: Adaptive Detector Operating-Point Control for Real-Time Aerial Multi-Object Tracking

Ahmed Gouda Ismail, Mohamed S. Mohamed, and Tarek Ahmed Mahmoud

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Computer Engineering and Artificial Intelligence Department, Military Technical College, Cairo, Egypt;

Computer Engineering Department, Egypt University of Informatics, Cairo, Egypt

Corresponding author: Ahmed Gouda Ismail, a7medgouda1@gmail.com

This supplementary document preserves supporting historical, implementation, and provenance evidence that is not required to understand the primary contribution or to verify the main-paper claims. Historical and modern protocols remain separate. All values below are copied from the frozen manuscript evidence and are not new experiments.

S1. Extended historical evidence

S1.1 Component ablation

The archived component progression is shown in full. The rows are historical evidence and are not pooled with our model’s results.

Table S1: Full historical component ablation.

Configuration	MOTA	HOTA	IDF1	IDS	FPS
OLD-A0	17.633	29.837	30.892	283	44.085
OLD-A1 tuned tracker	17.802	30.864	32.854	217	44.511
OLD-A2 confidence/NMS	17.636	31.384	33.727	210	40.651
OLD-A2R resolution	18.425	32.446	35.669	241	39.562
OLD-A3 full AC-MOT	18.165	33.064	36.296	271	37.962

S1.2 Historical VisDrone profiles

The historical frozen comparison contains separate operating profiles. V1 is quality-oriented, whereas V2 deliberately trades quality for identity and runtime behaviour.

Table S2: Full historical VisDrone comparison.

System	MOTA	HOTA	IDF1	IDS	FPS
Baseline	19.729	28.430	32.724	1235	36.528
Old AC-MOT	23.236	32.698	39.516	1061	41.957
V1 Trial 24	26.948	33.835	41.546	1184	38.985
V2 Trial 22	23.792	31.218	37.870	919	46.024

Table S3: Full archived V1 paired bootstrap against the historical baseline.

Metric	Difference	95% CI
MOTA	+7.2195	[5.4362, 9.2934]
HOTA	+5.4051	[4.1273, 6.9022]
IDF1	+8.8220	[6.9005, 11.0537]
IDS	–	not significant

Table S4: Full historical zero-retuning UAVDT transfer.

System	MOTA	HOTA	IDF1	IDS
Baseline	13.841	24.085	27.887	558
V1	17.399	28.390	34.453	321
V2	16.118	26.930	32.014	308

S1.3 Historical V1 uncertainty and UAVDT transfer

S1.4 V1 optimization record

The first formal optimization-derived configuration used fixed pretrained YOLOv8n and tuned ByteTrack on VisDrone2019-MOT-val. Test data were not used during search. The canonical Optuna/TPE stage searched five non-negative SCI weights normalized to sum one, confidence and NMS endpoints selected from the supported validation sets, and two sorted SCI thresholds. The preceding validation screens supplied resolution levels 512, 928, and 960; the temporal design supplied a seven-entry history and stride 10. The final directions were maximize MOTA, minimize IDS, and maximize FPS, followed by the validation-only feasibility rule FPS gate and IDS no greater than the OLD-A3 reference, then highest MOTA with the documented tie breaks. Trial24 was frozen.

The exact V1 weights are crowd 0.1294928, tiny 0.2217477, edge 0.4337134, night 0.0535577, and blur 0.1614885. The confidence endpoints are 0.30 and 0.40, both NMS endpoints are 0.35, and the SCI thresholds are 0.1353494 and 0.2872868. The V1 held-out bootstrap intervals and complete historical rows are in Tables S2 and S3; the values are not a new experiment.

S1.5 V2 optimization record

V2 reused the V1 validation design and changed the explicit objective to maximize MOTA and minimize IDS with only $\text{FPS} \geq 25$ as a feasibility gate. The implementation constructs the feasible Pareto front by the standard no-worse-in-both-objectives and strictly-better-in-one rule, then records highest-MOTA, lowest-IDS, and balanced candidates. The balanced candidate uses the documented equal-weight normalized MOTA/IDS score with HOTA, IDF1, FPS, and trial-index tie breaks. Trial22 was the official selected configuration; 49 trials (0–48) completed in the archived study. Its validation record is MOTA 19.330, HOTA 31.651, IDF1 34.226, IDS 168, and 51.406 FPS. The separate post-selection testdev rerun is 23.792 MOTA, 31.218 HOTA, 37.870 IDF1, 919 IDS, and 46.024 FPS, and is not the selection record.

S1.6 Cross-pipeline audit

The U2MOT host is the published ICCV implementation with YOLOX-X. V1 cue weights and transforms were retained, while U2MOT validation distributions provided unsupervised q33/q67 boundaries because the V1 boundaries collapsed all frames into the hard region. The held-out custom evaluator reported baseline/controller MOTA 53.9/53.7668, IDF1 69.8/69.8494, and IDS 1239/1152; no HOTA, confidence interval, or matched speed claim is available.

The SparseTrack host was evaluated on MOT17 val_half. Its transfer changes only NMS using an edge cue sampled every ten frames, with a threshold learned by leave-one-sequence-out development on the same seven sequences. Static and adaptive MOTA are 76.8881 and 76.9252, with bootstrap CI

$[-0.041, +0.182]$ for the change; the manifest does not establish superiority.

S2. Extended modern result evidence

S2.1 Full paired bootstrap metrics

The main paper reports the headline HOTA, MOTA, IDF1, and IDS intervals. The complete archived table additionally records false-positive and false-negative changes.

Table S5: Complete modern paired bootstrap changes versus the declared baseline.

Host	Metric	Difference	CI low	CI high
ByteTrack	HOTA	+2.937	+2.356	+3.375
ByteTrack	MOTA	+17.995	+13.541	+23.282
ByteTrack	IDF1	+5.360	+4.490	+6.178
ByteTrack	IDS	-325	-498	-164
ByteTrack	FP	-14382	-19176	-9620
ByteTrack	FN	+1781	+288	+3594.075
OATrack	HOTA	+1.325	+0.200	+2.264
OATrack	MOTA	+6.318	+4.741	+7.879
OATrack	IDF1	+2.258	+0.548	+3.474
OATrack	IDS	-169	-292	-42
OATrack	FP	-3994	-6190	-2494.900
OATrack	FN	-375	-1462.575	+670

The bootstrap uses 5000 paired sequence resamples with seed 42. It quantifies uncertainty across the available sequences; it does not make frames or detector stochasticity independent.

S2.2 State occupancy

Table S6: Our model’s state occupancy on the second VisDrone validation read.

Host	LOW	MEDIUM	HIGH
ByteTrack	7.38%	30.15%	62.47%
OATrack	7.38%	33.31%	59.31%

Occupancy verifies that the controller uses multiple states and that the trajectory differs by tracker host. It does not establish that exact switching timing causes the quality difference.

S2.3 Full modern UAVDT transfer

Table S7: Full modern UAVDT transfer under the frozen policy and no retuning.

Host	Policy	HOTA	MOTA	IDF1	IDS	FP	FN	Precision	Recall
ByteTrack	baseline	45.923	-1.573	58.400	601	279616	66050	49.571	80.625
ByteTrack	our model	47.004	16.028	61.767	486	212077	73702	55.751	78.381
OATrack	baseline	47.728	19.607	62.101	517	197937	75612	57.270	77.820
OATrack	our model	47.640	22.873	62.931	486	183669	78774	58.800	76.893

The transfer contains 20 sequences, 16,592 frames, and 49,776 detector forward passes. It is a single frozen-policy run per host without bootstrap or cross-validation.

S3. Implementation traces

S3.1 Original AC–MOT execution order

The early heuristic prototype is retained here as provenance and reproducibility evidence, not as the primary AC–MOT generation. The controller sits between a frame stream and a fixed detector–tracker pair. It reads cheap image statistics and boxes retained by the preceding tracker, computes a scene state, maps that state to detector confidence, NMS, and resolution, runs the detector, and stores the new tracker output for the next decision. It never reads future frames, ground truth, or the current action’s post-association output.

The historical cues are crowd count $C_t = \min(N_t/30, 1)$, the fraction of previous boxes below 32×32 pixels, normalized Canny edge density, a brightness-below-80 night indicator, and a Laplacian-variance-below-180 blur indicator. The historical SCI was

$$S_t = 0.30C_t + 0.30T_t + 0.20E_t + 0.10N_t + 0.05B_t,$$

whose coefficients intentionally sum to 0.95. The Smart Calibrator used $c_t = 0.245 - 0.050S_t$, applied a documented -0.012 correction for crowded, tiny, or night scenes, and clipped confidence to $[0.19, 0.28]$. It used $\eta_t = 0.490 - 0.050S_t$, applied a -0.012 blur correction, and clipped NMS to $[0.40, 0.52]$. The historical resolution actions were 640, 736, and 832 pixels, with analysis stride 10 and a seven-entry moving mean. These details are preserved to prevent the heuristic prototype from being confused with V1. The 832-pixel action was used when $S_t > 0.60$ or the tiny-object ratio exceeded 0.50; the 736-pixel action was used when $S_t > 0.35$ or the scene was labelled crowded or tiny; otherwise the action was 640 pixels. At an analysis frame, crowd and tiny-object values came from the previously stored tracker boxes, whereas edge, brightness, and blur came from the current reduced image. New boxes were stored only after detector execution and association, so the controller did not create a within-frame feedback loop. An empty previous output gave a zero tiny ratio and a crowd value that saturated at one after thirty boxes. Scene labels were evaluated alongside the scalar index, and the seven-entry mean reduced one-frame noise at the cost of response delay.

```
1: initialize seven-entry SCI history and previous tracker boxes
2: for each frame  $t$  do
3:   if  $t = 1$  or  $t \bmod 10 = 1$  then
4:     compute  $C_t, T_t, E_t, N_t, B_t$  from current image and prior boxes
5:     append  $S_t$  to the history and average the available entries
6:     apply Smart Calibrator confidence, NMS, and resolution rules
7:   end if
8:   run detector at held operating point and update the fixed tracker
9:   store tracker boxes for future analysis
10: end for
```

Algorithm S1: Original AC–MOT controller

S3.2 Execution trace for our model

At an analysis step, `analyze_visual` computes reduced grayscale statistics from the current image: Canny edge fraction, mean brightness, and Laplacian variance. The SceneLayer reads `prev_boxes`, reported by the tracker on the last processed frame, and computes box count, crowd value, box-area tiny fraction, and the decaying object-count peak. On non-analysis frames the previous state and action remain unchanged.

The implementation then evaluates the labels in order. Brightness can label a frame as night, blur can label it as blur, tiny-object fraction can label it as tiny, and crowd or edge conditions can label it as crowded. The raw target uses the modern thresholds and tiny guard. If the count falls below 0.40 of the decayed peak after the peak has reached five objects, the bounded recovery probe can request HIGH for up to three analysis steps.

If the raw target is lower than the persistent state, the hold rules are checked: HIGH is held when SCI remains above 0.50 or tiny remains above 0.40, and MEDIUM is held when SCI remains above 0.25. The 30-frame dwell gate then prevents a state change too soon after the previous change. Only after these guards does the adapter translate the state into the frozen resolution/NMS/confidence tuple, run the detector, update the tracker, and store its boxes for a later decision.

S3.3 Reproducibility checklist

1. Use the detector checkpoint `dronefreak` / `visdrone-yolo11m`.
2. Keep the eight-sequence calibration split separate from the seven-sequence modern evaluation read.
3. Preserve the crowd history length of seven and analysis stride of ten.
4. Apply the exact thresholds used by our model, including the tiny guard, recovery probe, hysteresis, 30-frame dwell, and action tuples.
5. Keep detector-side confidence distinct from OATrack’s tracker-side `min_conf=0.40`.
6. Use the class-agnostic project evaluator and label it as such rather than as an official VisDrone leaderboard protocol.
7. Pair the baseline and our model by sequence and use 5000 resamples with seed 42 for the modern bootstrap.
8. Do not retune on the seven-sequence read or on UAVDT.

S4. Provenance and selection audit

The modern freeze records the detector hash, controller thresholds, action map, tracker descriptions, random seed, calibration fold membership, forward-pass count, and source-file checksums. Stage 3A selected the cue subset [`'crowd'`]; this describes the weighted SCI, not the removal of tiny, edge, brightness, or blur from the surrounding SceneLayer.

The seven-sequence modern VisDrone evaluation is a documented second validation read. It must not be described as an untouched held-out confirmation. The archive also records that frozen Trial 7 is not the literal argmax of the documented selection rule under either archived eligibility interpretation. The manuscript reports this deviation and does not rerun the search.

The matched static control uses 1088 resolution, NMS IoU 0.45, and confidence 0.40. Its HOTA is approximately 44.530 for ByteTrack and 48.269 for OATrack, compared with our model’s values 44.523 and 48.095. The shuffled timing control has development mean ΔHOTA 2.446957 for the causal schedule and 2.617228 for the same-action-distribution shuffled schedule. These controls shift the safe interpretation toward operating-point calibration and observation-set shaping; they do not establish that random timing is globally superior.

S5. Boundary of the supplementary evidence

The supplementary material supports auditability and future reproduction. It does not add a new claim of universal tracker independence, official leaderboard equivalence, 30-FPS operation, or a timing-specific causal gain. The main paper retains the negative attribution evidence, the second-read disclosure, the Trial 7 selection deviation, the runtime harness caveat, and the mixed OATrack UAVDT transfer because those facts define the supported claim boundary.